

Chapter 7 – Seasonal ARIMA Models

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1 Introduction

1. In this lecture, we look at time series data that exhibit seasonality.
2. A lot of time series data is collected on a quarterly basis, or on a monthly basis.
3. Data such as monthly rainfall or monthly temperature or quarterly revenues etc. normally will exhibit seasonality.
4. How do we work with such data?
5. Firstly, we do a plot to see if there is seasonality. If there is, take note of the number of periods s (called the seasons) where the seasonal pattern repeats itself. So from now on, we assume that we are working with a time series where we have already established that there is a seasonal pattern with seasons s .
6. Next if the trend looks exponentially increasing, we take log transformation first and work with $\log x_t$. Note that $\log x_t$ will also be seasonal with seasons s . Then we need to look at the correlogram and partial correlogram of

$$(1 - B^s) \log x_t = \log x_t - \log x_{t-s}.$$

Hopefully this will give us a suitable ARMA model for $(1 - B^s) \log x_t$.

7. If there is a linear instead of an exponential trend, then we look at

$$\nabla(1 - B^s)x_t = (x_t - x_{t-s}) - (x_{t-1} - x_{t-s-1})$$

and its correlogram and partial correlogram to see if we can fit an ARIMA model to $(1 - B^s)x_t$ first.

8. Suppose we take $\log x_t$, and then it still looks like there is still a linear trend, then we need to look at $\nabla \log x_t$ and assuming that the mean now looks to be around zero except for the seasonality, then we look at the correlogram and partial coorelograms of

$$\begin{aligned} (1 - B^s)\nabla \log x_t &= (1 - B^s)(\log x_t - \log x_{t-1}) \\ &= (\log x_t - \log x_{t-1}) - (\log x_{t-s} - \log x_{t-s-1}) \end{aligned}$$

Hopefully this will give us a suitable ARIMA model for $(1 - B^s) \log x_t$

9. In practice, most of the correlogram and partial correlogram plots will not look clear-cut and so we proceed by increasing the order of the terms one by one and check for the lowest AIC value and also be guided by our appetite for the complexity of the model, e.g., is it worthwhile to have a model with over 10 terms that has the minimum AIC value compared to one with 5 terms but with a slightly higher AIC value.

2 The General SARIMA Model

1. The general seasonal ARIMA model is also called the SARIMA model.
2. The SARIMA model uses differencing at a lag equal to the number of seasons s .
3. In the earlier lectures, we used lag 1 differencing to remove a linear trend, that is, using $\nabla x_t = (1 - B)x_t$.
4. So to remove seasons, we use $(1 - B^s)x_t$, that is why we try to look at the correlogram and the partial correlogram of $(1 - B^s)x_t$.
5. The general SARIMA model includes AR and MA terms at lag s , but also includes other AR and MA terms at smaller lags, since time series observations of seasonal data may also depend on lags that are immediately closer to those particular observation.
6. The general SARIMA model denoted by $\text{ARIMA}(p, d, q)(P, D, Q)_s$ is given by

$$\Theta_P(B^s)\theta_p(B)(1 - B^s)^D(1 - B)^d x_t = \Phi_Q(B^s)\phi_q(B)w_t$$

where Θ_P , θ_p , Φ_Q and ϕ_q are polynomials of orders P , p , Q and q respectively.

7. Note that $\Theta_P(B^s)$ and $\Phi_Q(B^s)$ will be polynomials in terms of powers of B^s .
8. These are some simpler models.
 - (a) A simple AR model with a seasonal period of 12 units where the current observation only depends on lag 12 is denoted by $\text{ARIMA}(0, 0, 0)(1, 0, 0)_{12}$ and takes the form

$$x_t = \alpha x_{t-12} + w_t.$$

The model is stationary when $|\alpha^{-1/12}| > 1$.

- (b) If there is also a linear trend, then the model in (a) can be extended to $\nabla x_t = \alpha \nabla x_{t-12} + w_t$ which simplifies into

$$x_t = x_{t-1} + \alpha x_{t-12} - \alpha x_{t-13} + w_t.$$

This can be factorised as

$$(1 - \alpha B^{12})(1 - B)x_t = w_t,$$

where comparing with item 2.6, tells us that this model is ARIMA(0, 1, 0)(1, 0, 0)₁₂. This model is non-stationary since $B = 1$ is one of the roots.

- (c) A simple quarterly seasonal moving average model is $x_t = (1 + \beta B^4)w_t = w_t + \beta w_{t-4}$. This is an ARIMA(0, 0, 0)(0, 0, 1)₄ model. If the data contains a linear trend, this can be extended to $x_t = x_{t-1} + w_t + \beta w_{t-4}$ which is $(1 - B)x_t = (1 + \beta B^4)w_t$ and thus this is ARIMA(0, 1, 0)(0, 0, 1)₄.

9. In general, for seasonal time series data where there are a lot of different things mixed up together, it is most of the time still difficult to figure out the significant lags in the partial correlogram of $(1 - B^s)x_t$, even if there are no trends, so we just try fitting the model by starting with small values of the parameters and keep increasing until we get the lowest AIC value.

3 Examples

1. **Example 1.** We look at the UKDriverDeaths data in **R** .

```
> plot(UKDriverDeaths)
> plot(diff(UKDriverDeaths,lag=12),type="l")
> mean(diff(UKDriverDeaths, lag=12))
[1] -19.61111
```

We do a plot of the raw data and see that it is clearly seasonal with $s = 12$ since this is monthly data. The second plot is actually a plot of $(1 - B^{12})x_t$ which is $x_t - x_{t-12}$ and this removes the seasonal effects and the resulting plot shows that $x_t - x_{t-12}$. The mean of $x_t - x_{t-12}$ is -19.61 but relative to the range of values of $x_t - x_{t-12}$, it appears to be insignificant, thus $x_t - x_{t-12}$ could very look like a stationary process, so we check the correlogram and partial correlogram of $x_t - x_{t-12}$.

```
> acf(diff(UKDriverDeaths,lag=12),lag.max=70)
> pacf(diff(UKDriverDeaths,lag=12),lag.max=70)
```

The correlogram of $x_t - x_{t-12}$ shows significance to lag 5 and then lag 12 the partial correlogram shows significance to lag 3, then 12, 13, then around 23,25.

We will try seasonal ARIMA(0,0,0)(1,1,0)₁₂ first, although this is most likely inadequate since the model equation will only contain terms involving x_{t-12} and x_{t-24} terms only.

```
> arima(ts(UKDriverDeaths),order=c(0,0,0),season=list(order=c(1,1,0),period=12))
```

Call:

```
arima(x = ts(UKDriverDeaths), order = c(0, 0, 0), seasonal = list(order = c(1,
  1, 0), period = 12))
```

Coefficients:

```
      sar1
    -0.2617
s.e.    0.0727
```

```
sigma^2 estimated as 37738:  log likelihood = -1204.29,  aic = 2412.58
```

Now we try

```
> arima(ts(UKDriverDeaths),order=c(1,0,0),season=list(order=c(1,1,0),period=12))
```

Call:

```
arima(x = ts(UKDriverDeaths), order = c(1, 0, 0), seasonal = list(order = c(1,
  1, 0), period = 12))
```

Coefficients:

	ar1	sar1
	0.5554	-0.4147
s.e.	0.0633	0.0686

sigma² estimated as 26467: log likelihood = -1173.25, aic = 2352.51

```
> arima(ts(UKDriverDeaths),order=c(2,0,0),season=list(order=c(1,1,0),period=12))
```

Call:

```
arima(x = ts(UKDriverDeaths), order = c(2, 0, 0), seasonal = list(order = c(1,
  1, 0), period = 12))
```

Coefficients:

	ar1	ar2	sar1
	0.3960	0.2947	-0.4411
s.e.	0.0711	0.0714	0.0680

sigma² estimated as 24121: log likelihood = -1165.14, aic = 2338.29

```
> arima(ts(UKDriverDeaths),order=c(3,0,0),season=list(order=c(1,1,0),period=12))
```

Call:

```
arima(x = ts(UKDriverDeaths), order = c(3, 0, 0), seasonal = list(order = c(1,
  1, 0), period = 12))
```

Coefficients:

	ar1	ar2	ar3	sar1
	0.3876	0.2836	0.0286	-0.4413
s.e.	0.0743	0.0770	0.0744	0.0679

sigma² estimated as 24102: log likelihood = -1165.07, aic = 2340.14

The second of these 3 models is seasonal ARIMA(2,0,0)(1,1,0)₁₂ and it appears that it has got the lowest AIC so far. We look at the correlogram of its residual.

```
> acf(arima(ts(UKDriverDeaths),order=c(3,0,0),season=list(order=c(1,1,0),period=12))$resid,lag.max=70)
> pacf(arima(ts(UKDriverDeaths),order=c(3,0,0),season=list(order=c(1,1,0),period=12))$resid,lag.max=70)
```

There are still some significance at lags 24 and 36, so any model might

need to include some terms like x_{t-24} and x_{t-36} and possibly any neighbouring terms around these.

After some trial and error, we seem to get the seasonal ARIMA(3, 1, 2)(2, 1, 1)₁₂ as the model that has the smallest AIC so far and also one that does not have any convergence problems in the calculations.

```
> arima(ts(UKDriverDeaths),order=c(3,1,2),season=list(order=c(2,1,1),period=12))
```

Call:

```
arima(x = ts(UKDriverDeaths), order = c(3, 1, 2), seasonal = list(order = c(2, 1, 1), period = 12))
```

Coefficients:

	ar1	ar2	ar3	ma1	ma2	sar1	sar2	sma1
	-0.6471	0.3894	0.1738	0.1296	-0.8550	0.0729	-0.0237	-0.9172
s.e.	0.1065	0.1459	0.0947	0.1161	0.1041	0.1006	0.0925	0.1364

sigma^2 estimated as 16515: log likelihood = -1135.79, aic = 2289.58

```
> acf(arima(ts(UKDriverDeaths),order=c(3,1,2),season=list(order=c(2,1,1),period=12))$resid, lag.max=70)
```

A look of the correlogram of the residuals show that all lags except 2 are insignificant and the two that are significant are only marginally so. Thus this model is

$$(1 - 0.0729B^{12} + 0.0237B^{24})(1 + 0.6471B - 0.3894B^2 - 0.1738B^3)(1 - B^{12})(1 - B)x_t \\ = (1 - 0.9172B^{12})(1 + 0.1296B - 0.8550B^2)w_t.$$

However, with further experimenting, we obtain a seasonal ARIMA(3, 0, 11)(2, 1, 1)₁₂ model which gives perfect white noise as the residuals.

```
> arima(UKDriverDeaths,order=c(3,0,11),season=list(order=c(2,1,1),period=12))
```

Call:

```
arima(x = UKDriverDeaths, order = c(3, 0, 11), seasonal = list(order = c(2, 1, 1), period = 12))
```

Coefficients:

	ar1	ar2	ar3	ma1	ma2	ma3	ma4	ma5	ma6	ma7
	-0.5105	0.5433	0.9658	1.0652	0.1121	-0.7700	-0.3253	-0.0552	0.0008	-0.0039
s.e.	0.0321	0.0281	0.0271	0.0816	0.1120	0.1124	0.1257	0.1325	0.1396	0.1247
	ma8	ma9	ma10	ma11	sar1	sar2	sma1			
	-0.2553	-0.1276	0.1208	0.2681	0.1334	-0.0456	-1.000			
s.e.	0.1316	0.1102	0.1197	0.0836	0.0906	0.0868	0.029			

sigma^2 estimated as 14636: log likelihood = -1134.69, aic = 2305.38

```
> acf(arima(UKDriverDeaths,order=c(3,0,11),season=list(order=c(2,1,1),period=12))$resid, lag.max=80)
```

```
> pacf(arima(UKDriverDeaths,order=c(3,0,11),season=list(order=c(2,1,1),period=12))$resid, lag.max=80)
```

The equation of this model is

$$(1 - 0.1334B^{12} + 0.0456B^{24})(1 + 0.5105B - 0.5433B^2 - 0.9658B^3)(1 - B^{12})x_t \\ = (1 - B^{12}) \left(1 + \sum_{i=1}^{11} \beta_i B^i \right) w_t.$$

So if we need to do forecasting, we can use the above equation to carry out the successive calculations for x_{t+1} , x_{t+2} and so on, but given the complicated algebra, it is rather tedious to do so, instead, there is another function `predict()`.

```
> predict(arima(UKDriverDeaths,order=c(3,0,10),season=list(order=c(2,1,1),period=12)), n.ahead=24)
$pred
      Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
1985 1378.375 1196.677 1246.502 1159.469 1242.237 1251.127 1292.423 1317.517 1392.986 1508.320
1986 1393.135 1214.397 1267.731 1143.408 1296.696 1242.896 1309.488 1344.448 1387.158 1526.850
      Nov      Dec
1985 1727.064 1852.510
1986 1741.932 1851.690

$se
      Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
1985 125.2178 139.5984 148.5615 153.0528 155.5087 162.2778 164.5863 170.1169 170.8360 174.8940
1986 185.7041 188.6476 190.6073 194.7080 196.5087 198.8029 202.0037 203.7086 205.8149 208.4854
      Nov      Dec
1985 178.1203 179.1144
1986 210.0064 212.1619

> plot(predict(arima(UKDriverDeaths,order=c(3,0,10),season=list(order=c(2,1,1),period=12)), n.ahead=24)$pred
```

Here we forecast the deaths for the next 24 months ahead and do a plot. Note that the plot of the forecast values also show the seasonality.

- Example 2.** We use the `USAccDeaths` data in **R**. My best fit appears to be a seasonal ARIMA(1, 1, 1)(1, 1, 1)₁₂ and the residuals of the model resemble that of white noise.

```
> plot(USAccDeaths)
> plot(diff(USAccDeaths, lag=12))
> plot(diff(diff(USAccDeaths, lag=12)))
> acf(diff(diff(USAccDeaths, lag=12)), lag.max = 35)
> pacf(diff(diff(USAccDeaths, lag=12)), lag.max = 35)
> arima(USAccDeaths,order=c(1,1,0),season=list(order=c(1,1,0),12))
```

Call:

```
arima(x = USAccDeaths, order = c(1, 1, 0), seasonal = list(order = c(1, 1, 0),
12))
```

Coefficients:

```
      ar1      sar1
-0.3325  -0.3515
```



```
s.e.    0.1214    0.1250
```

```
sigma^2 estimated as 116222:  log likelihood = -428.63,  aic = 863.27
> arima(USAccDeaths,order=c(1,1,1),season=list(order=c(1,1,1),12))
```

Call:

```
arima(x = USAccDeaths, order = c(1, 1, 1), seasonal = list(order = c(1, 1, 1),
  12))
```

Coefficients:

```
          ar1          ma1          sar1          sma1
          0.0762   -0.4867    0.3004   -0.9940
s.e.    0.2967    0.2601    0.1776    1.2269
```

```
sigma^2 estimated as 81497:  log likelihood = -424.99,  aic = 859.98
```

```
> acf(arima(USAccDeaths,order=c(1,1,1),season=list(order=c(1,1,1),12))$resid,lag.max=30)
```

Thus this model is

$$(1 - 0.3004B^{12})(1 - 0.0762B)(1 - B^{12})(1 - B)x_t \\ = (1 - 0.9940B^{12})(1 - 0.4867B)w_t.$$

If we are forecasting ahead for the next 12 months, then

```
> predict(arima(USAccDeaths,order=c(1,1,1),season=list(order=c(1,1,1),12)),n.ahead=12)
$pred
      Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct
1979 8256.991 7417.119 8238.220 8515.716 9391.549 9825.008 10732.763 10050.862 9117.060 9290.515
      Nov      Dec
1979 8786.053 9140.554

$se
      Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct      Nov
1979 307.3671 355.9924 394.5295 429.3364 461.5054 491.5722 519.9031 546.7679 572.3732 596.8813 620.4259
      Dec
1979 643.1558
```